20260311_BEL2_Math

Complex Numbers.

We will learn

- rectangular form $a+jb$
- geometric meaning
- polar form $re^{j\theta}$

and we will see why this matters for

- frequency analysis
- differential equations
- Laplace transform

$\boxed{x^2+1=0}$ Can I solve this eq. in $\mathbb{R}$?

In $\mathbb{R}$

$x^2 \geq 0 \quad \forall x \in \mathbb{R}$ ✓

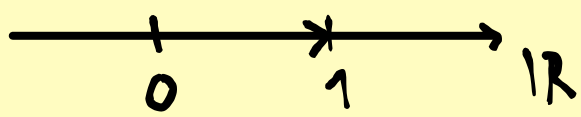
$x^2+1 \geq 1 \quad \not\Leftrightarrow \quad x^2+1=0$

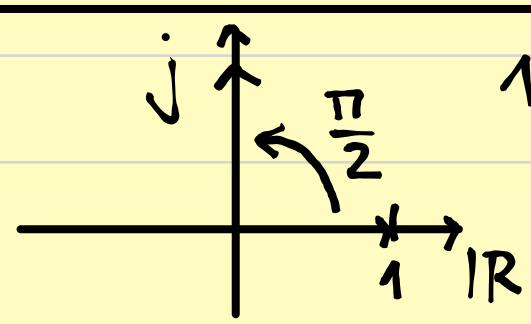
We introduce a new number $\boxed{j}$ such that $j^2=-1$.

$j^2+1=0 \rightarrow -1+1=0 \rightarrow j$ is a solution to $x^2+1=0$

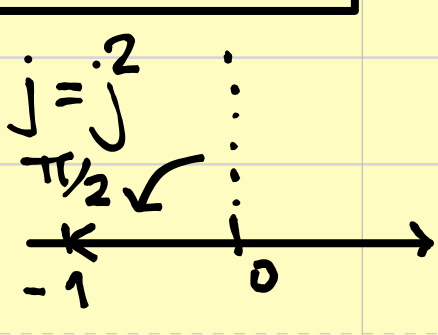
$\boxed{\text{What is the intuitive, geometric meaning of } j^2?}$

$1 \cdot j$





$1 \cdot j \cdot j = j^2$



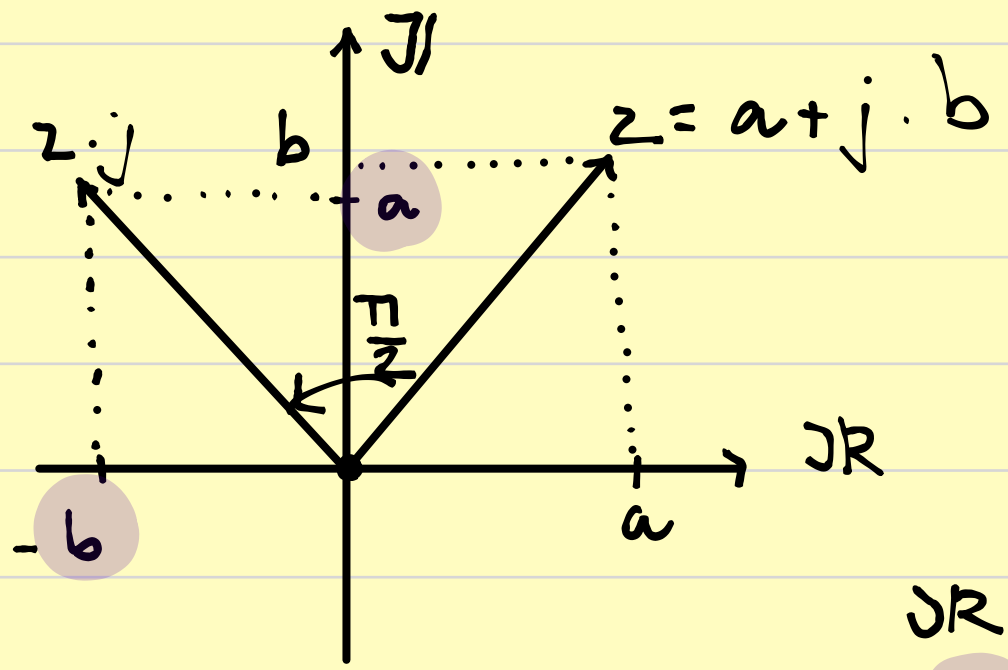
A complex number is a number of the form

$$\boxed{z \in \mathbb{C} \qquad z = a + j \cdot b \qquad a, b \in \mathbb{R}}$$

- a is the "real part of z" $\quad \Re(z) = a$
- b is the "imaginary part of z" $\quad \Im(z) = b$

$$\boxed{\text{Every real number is a complex number with } b = 0.}$$

$\mathbb{R} \subset \mathbb{C}$



example:

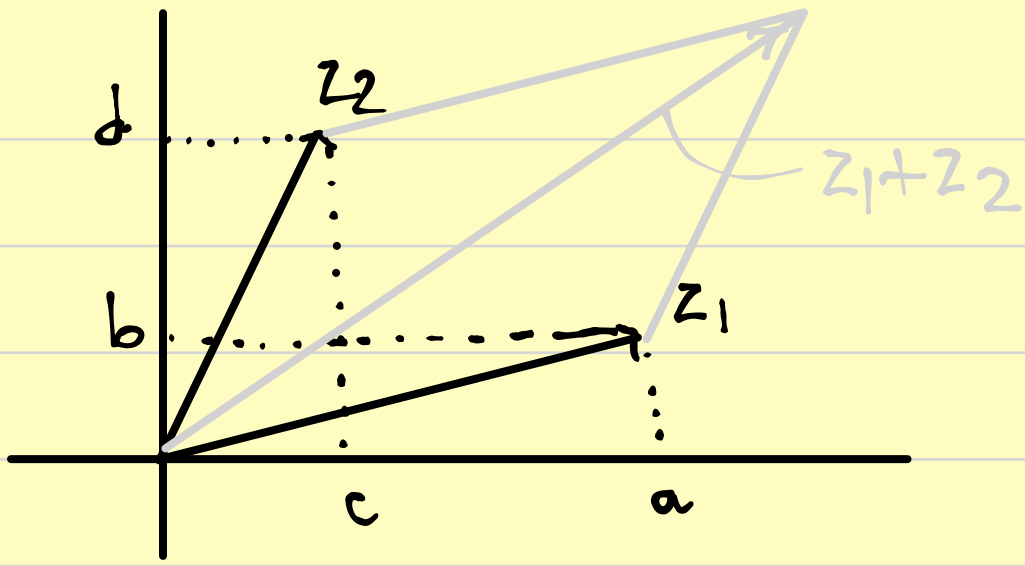
$$z \cdot j = (a + j \cdot b) \cdot j = a \cdot j + j^2 \cdot b = \underbrace{-b}_{\Re} + j \cdot \underbrace{a}_{\Im}$$

example:

$$z_1 = a + j \cdot b \qquad z_2 = c + j\, d \qquad a, b, c, d \in \mathbb{R}$$

$$\boxed{\text{EQUALITY: } \quad a + j \cdot b = c + j \cdot d \underset{\text{iff}}{\longleftrightarrow} a = c \ \& \ b = d}$$

$$\boxed{\text{ADDITION: } (a + jb) + (c + j \cdot d) = (a + c) + j(b + d)}$$



MULTIPLICATION

$$z_1 = a + jb \qquad z_2 = c + jd$$

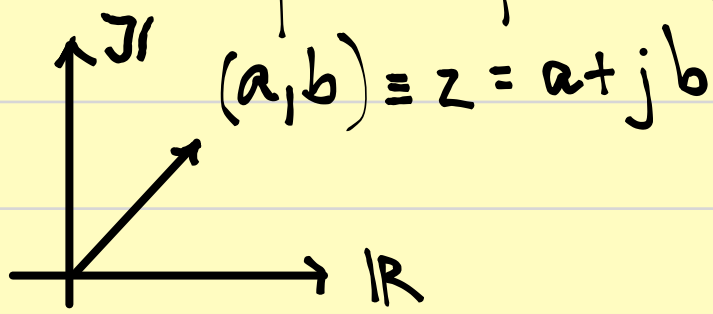
$$z_1 \cdot z_2 = (a+jb)(c+jd) = a \cdot c + j^2 b \cdot d + j(b \cdot c + a \cdot d)$$

$$= (a \cdot c - b \cdot d) + j(b \cdot c + a \cdot d)$$

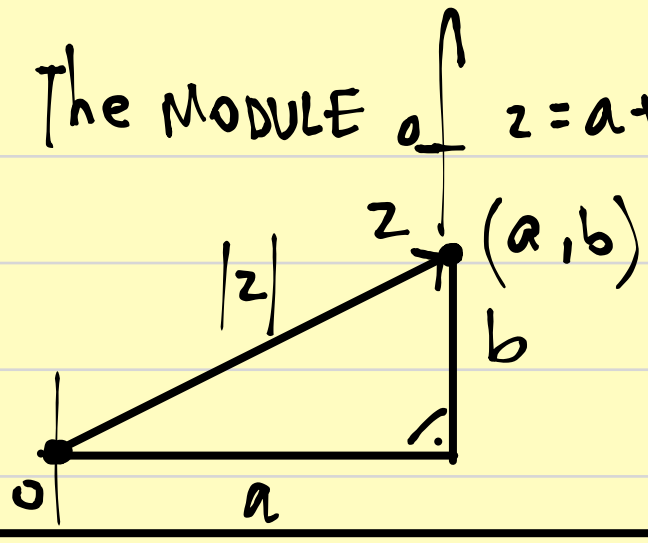
example:

- $(1+2j)+(3-j) = 1+3+(2-1)j = 4+j$
- $(1+2j)(3-j) = 3-j+6j+2 = 5+5j$

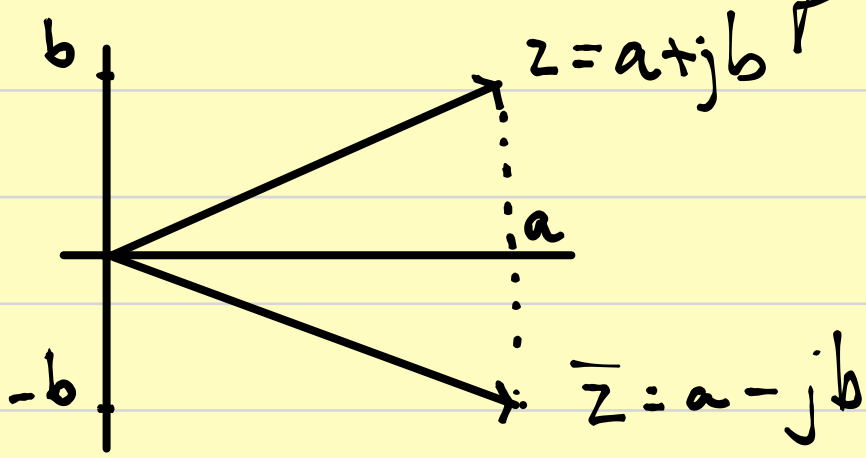
A complex number $z = a+jb$ can be represented by the point (a,b) in the complex plane.



The MODULE of $z = a + jb$ is $|z| = \sqrt{a^2 + b^2}$



The COMPLEX CONJUGATE of $z = a + jb$ is $\bar{z} = a - jb$



$$z \cdot \bar{z} = (a + jb)(a - jb) = a^2 - \cancel{jab} + \cancel{jba} - j^2 b^2$$

$$= a^2 + b^2 = |z|^2$$

$$\boxed{z \cdot \bar{z} = |z|^2}$$

DIVISION

$$\frac{z_1}{z_2} = \frac{a + jb}{c + jd}$$

Denominator becomes $\mathbb{R}$

$$\frac{z_1}{z_2} = \frac{a + jb}{c + jd} \cdot \frac{c - j \cdot d}{c - j \cdot d} = \frac{a \cdot c + bd + j(b \cdot c - a \cdot d)}{c^2 + d^2}$$

example: $\dfrac{2 + 3j}{1 - j} = \dfrac{(2 + 3j)(1 + j)}{1^2 + 1^2} = \dfrac{-1 + 5j}{2}$

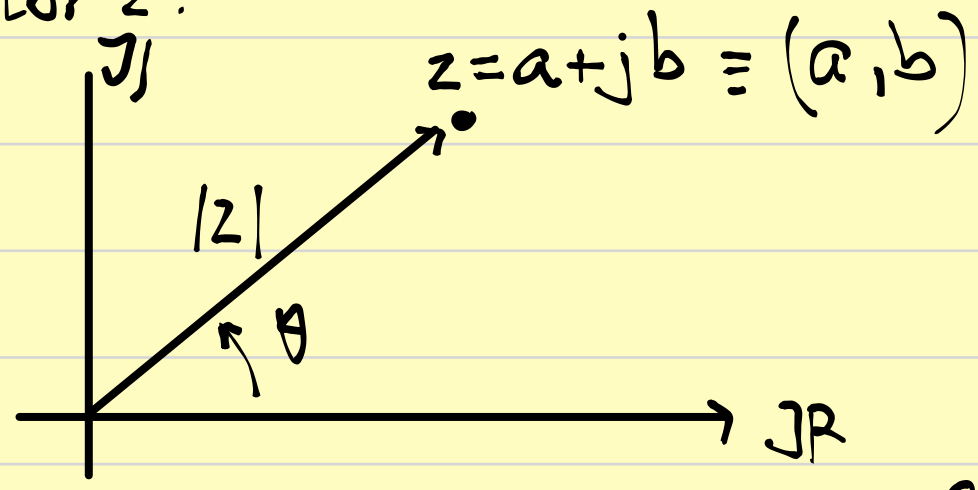
Polar form

Is best for multiplication, division and powers.

$z = a + jb \neq 0$

Let $r = |z| = \sqrt{a^2+b^2}$

Let θ be the angle btw the positive real axis and the vector z.



$$z = r \cdot \cos\theta + j \cdot r \sin\theta = r e^{j\theta}$$

Angles are not unique! $\theta, \theta + 2\pi, \theta + 4\pi, \ldots$

Sometimes students use $\theta = \tan^{-1}\left(\frac{b}{a}\right)$.
That is dangerous because it might give you the wrong quadrant.

ALWAYS check the sign of the real part!

example. $z = 1 - j$ $\quad r = \sqrt{1^2 + (-1)^2} = \sqrt{2}$

$$\theta = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4} \qquad 1 > 0!$$

$$\boxed{\text{EULER'S FORMULA} \quad e^{j\theta} = \cos\theta + j\sin\theta}$$

$$\boxed{z = re^{i\phi} \qquad \text{POLAR FORM}}$$

$$z_1 = r_1 e^{j\theta_1} \qquad z_2 = r_2 e^{j\phi_2}$$

MULTIPLICATION $\quad z_1 z_2 = r_1 r_2 e^{j(\theta_1+\theta_2)}$

DIVISION $\quad \dfrac{z_1}{z_2} = \dfrac{r_1}{r_2} e^{j(\theta_1-\theta_2)}$

Geometric interpretation:

- Multiplying by $e^{j\theta}$ means ROTATE by angle θ
- Multiplying by r means SCALE by factor r